Program:

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error
iris = load_iris()
data = pd.DataFrame(data=iris.data, columns=iris.feature_names)
data['species'] = iris.target
print(data.head())
X = data[['sepal length (cm)']]
y = data['sepal width (cm)']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3,
random_state=42)
model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
mse = mean_squared_error(y_test, y_pred)
print(f'Mean Squared Error on test set: {mse:.2f}')
plt.figure(figsize=(10, 6))
plt.scatter(X_test, y_test, color='r',marker='*', label='Actual')
plt.plot(X_test, y_pred, color='y', linewidth=3, label='Predicted')
plt.xlabel('Sepal Length (cm)')
plt.ylabel('Sepal Width (cm)')
plt.title('Linear Regression: Sepal Width vs Sepal Length')
plt.legend()
plt.show()
```

```
new_sample = pd.DataFrame([[5]], columns=['sepal length (cm)'])

predicted_width = model.predict(new_sample)

print(f'The predicted sepal width for sepal length {new_sample.values.tolist()} is {predicted_width[0]:.2f} cm')
```

Output:

```
   sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)  \
0                5.1               3.5                1.4               0.2
1                4.9               3.0                1.4               0.2
2                4.7               3.2                1.3               0.2
3                4.6               3.1                1.5               0.2
4                5.0               3.6                1.4               0.2

   species
0        0
1        0
2        0
3        0
4        0
Mean Squared Error on test set: 0.23
```